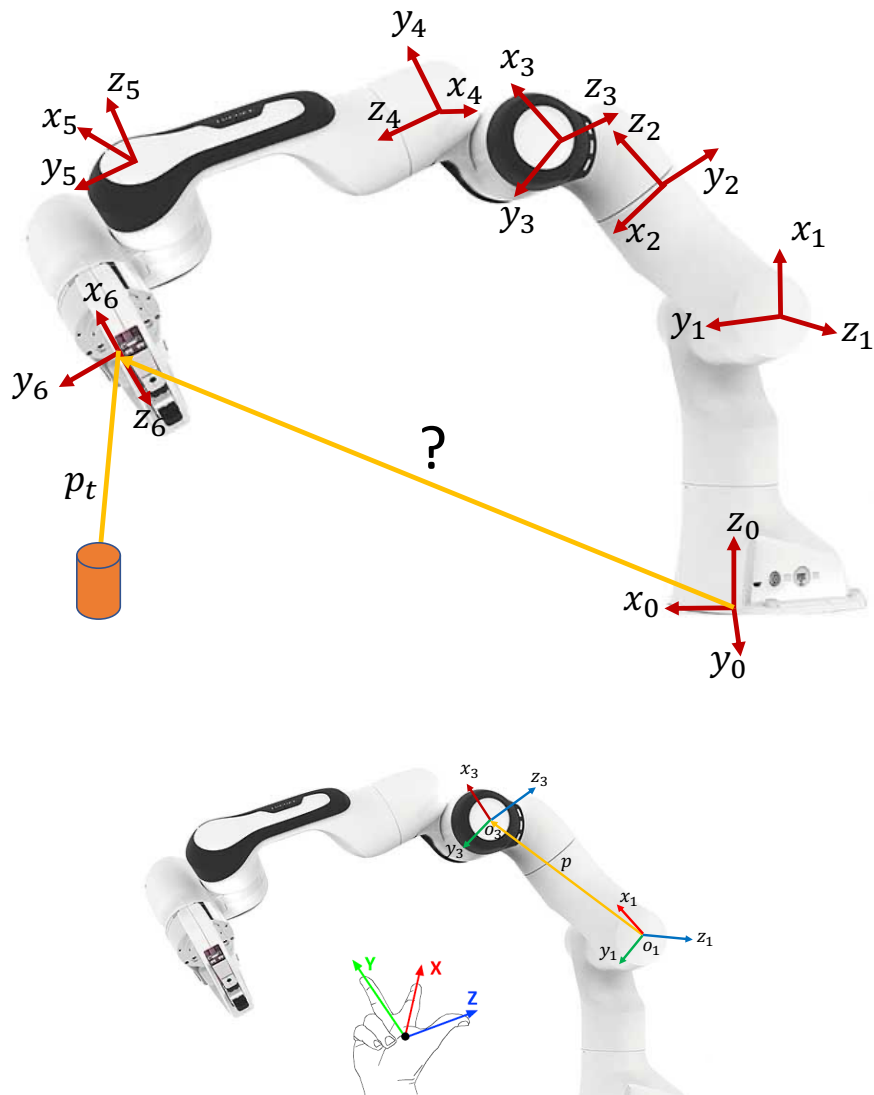


3D Kinematics tutorial-1

Learning outcomes:

- Know what the forward kinematics problem is
- Know how to assign frames to joints in a 3D robot.
- Know what a homogeneous coordinate system is.
- Know how to interpret a homogeneous transformation matrix.

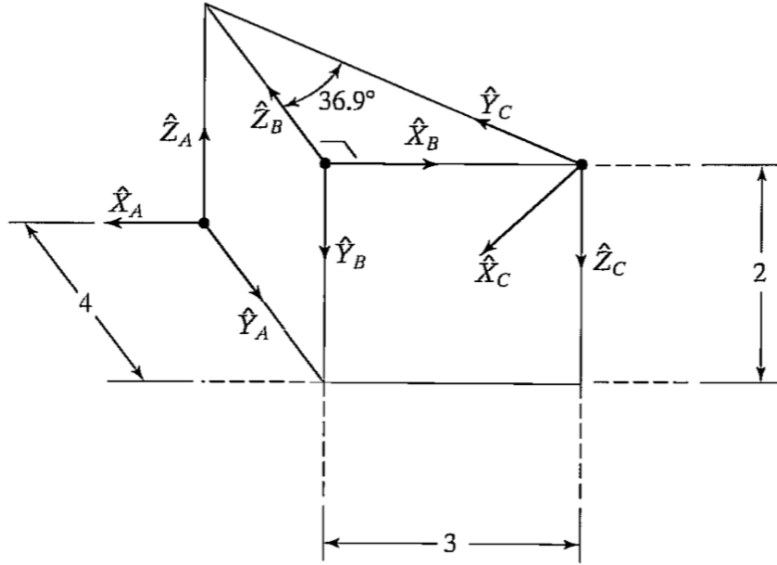
The forward kinematics problem Given knowledge of only the joint angles and link lengths, how to determine the position and orientation of the end-effector of a robot relative to the base.



A coordinate frame i consists of an origin, denoted O_i , and three mutually orthogonal basis vectors, denoted (x_i, y_i, z_i) , that are fixed within a given body.

Question-1

Craig, Q 2.32 Find the transformation matrices ${}^B T_C$, ${}^A T_B$, and ${}^A T_C$.



Question-2

You found a piece of paper with a set of rotation matrices. However, some of the numbers are not clear. Which of the following rotation matrices could be the complete set?

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ ? & ? & ? \end{bmatrix} \quad \begin{bmatrix} ? & 0 & 0.7071 \\ ? & 1 & 0 \\ ? & 0 & 0.7071 \end{bmatrix} \quad \begin{bmatrix} 0.866 & ? & 0 \\ 0.5 & ? & 0 \\ 0 & ? & 1 \end{bmatrix}$$

a)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0.7071 & 0 & 0.7071 \\ 0 & 1 & 0 \\ -0.7071 & 0 & 0.7071 \end{bmatrix} \quad \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

b)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0.7071 & 0 & 0.7071 \\ 0 & 1 & 0 \\ 0 & 0 & 0.7071 \end{bmatrix} \quad \begin{bmatrix} 0.866 & 0 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

c)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix} \quad \begin{bmatrix} -0.7071 & 0 & 0.7071 \\ 0 & 1 & 0 \\ -0.7071 & 0 & 0.7071 \end{bmatrix} \quad \begin{bmatrix} 0.866 & 0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

d)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \quad \begin{bmatrix} -0.7071 & 0 & 0.7071 \\ 0 & 1 & 0 \\ 1 & 0 & 0.7071 \end{bmatrix} \quad \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & -0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Question-3

A 3D rotation matrix is given by $\mathbf{R}_{3 \times 3}$. Given the transformation $\begin{bmatrix} \hat{\mathbf{v}}_{3 \times 1} \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{3 \times 3} & \mathbf{0}_{3 \times 1} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{v}_{3 \times 1} \\ 1 \end{bmatrix}$

what is the value of the ratio given by $\frac{\hat{v}}{v}$, where $\mathbf{x}$ is the size of the vector $\mathbf{x}$? Explain your answer.

Choices:

- 1) Not enough information
- 2) 1
- 3) Less than 1
- 4) Greater than 1

Question-4

In an $X - Y - Z$ Euler rotation, after doing a rotation around X , the new Y axis is Y' and the new Z axis is Z' . After doing a second rotation around Y' , the new X axis is X'' and the new Z axis is Z'' . There is a vector v such that $v = k_1 X + k_2 Y'$, and k_1, k_2 are real numbers. Is a third rotation around v a valid Euler rotation? Explain your answer.

Choices:

- 1) No
- 2) Yes

Question-5

Which of the following give the correct expressions for rotation angles given the following rotation matrix?

$${}^i R_j(\alpha, \beta, \gamma) = \begin{bmatrix} c\alpha c\beta c\gamma - s\alpha s\gamma & -c\alpha c\beta s\gamma - s\alpha c\gamma & c\alpha s\beta \\ s\alpha c\beta c\gamma - c\alpha s\gamma & -s\alpha c\beta s\gamma + c\alpha c\gamma & s\alpha s\beta \\ -s\beta c\gamma & s\beta s\gamma & c\beta \end{bmatrix} \quad (1)$$

$$= \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

a)

$$\beta = \text{atan2}(\sqrt{r_{33}^2 + r_{32}^2}, r_{31}); \alpha = \text{atan2}(\frac{r_{32}}{s\beta}, -\frac{r_{31}}{s\beta}); \gamma = \text{atan2}(\frac{r_{23}}{s\beta}, \frac{r_{13}}{s\beta}) \quad (2)$$

b)

$$\beta = \text{atan2}(r_{31}, \sqrt{r_{32}^2 + r_{33}^2}); \alpha = \text{atan2}(\frac{r_{21}}{c\beta}, \frac{r_{11}}{c\beta}); \gamma = \text{atan2}(\frac{r_{32}}{c\beta}, \frac{r_{33}}{c\beta}) \quad (3)$$

c)

$$\beta = \text{atan2}(r_{31}, \sqrt{r_{32}^2 + r_{33}^2}); \alpha = \text{atan2}(\frac{r_{23}}{s\beta}, \frac{r_{13}}{s\beta}); \gamma = \text{atan2}(\frac{r_{32}}{s\beta}, -\frac{r_{31}}{s\beta}) \quad (4)$$

d)

$$\beta = \text{atan2}(\sqrt{r_{31}^2 + r_{32}^2}, r_{33}); \alpha = \text{atan2}(\frac{r_{23}}{s\beta}, \frac{r_{13}}{s\beta}); \gamma = \text{atan2}(\frac{r_{32}}{s\beta}, -\frac{r_{31}}{s\beta}) \quad (5)$$

Question-6

"A rotation of a frame in 3D space around an axis passing through its origin can be realized through 3 successive rotations around any 3 linearly independent axes". Is this statement correct?

a) Correct

b) Not correct

Question-7

Which of the following are NOT true about a 3×3 rotation matrix R ?

a) The determinant of R is always 1

b) The sum of the diagonal of $R^T R$ is 3×3 rotation matrix

c) $R^{-1} = R^T$

d) $R^T R^{-1} = I$, where I is an identity matrix.

Question-8

A vector in Euclidean representation is given by

$$v = \begin{bmatrix} 0.5 \\ 0.25 \\ 0.75 \end{bmatrix} \quad (6)$$

What would be its homogeneous coordinate representation?

a)

$$v = \begin{bmatrix} 1 \\ 0.5 \\ 1.5 \end{bmatrix} \quad (7)$$

b)

$$v = \begin{bmatrix} 1 \\ 0.5 \\ 1.5 \\ 1 \end{bmatrix} \quad (8)$$

c)

$$v = \begin{bmatrix} 1 \\ 0.5 \\ 1.5 \\ 2 \end{bmatrix} \quad (9)$$

d)

$$v = \begin{bmatrix} 0.5 \\ 0.25 \\ 0.75 \\ 1 \end{bmatrix} \quad (10)$$

Question-9

Four homogeneous transformation matrices, $T_1 = \begin{bmatrix} R_1 & P_1 \\ o & 1 \end{bmatrix}$, $T_2 = \begin{bmatrix} R_2 & P_2 \\ o & 1 \end{bmatrix}$, $T_3 = \begin{bmatrix} R_3 & P_3 \\ o & 1 \end{bmatrix}$, $T_4 = \begin{bmatrix} R_4 & P_4 \\ o & 1 \end{bmatrix}$, are related as given by $T_1 T_2 T_3 = T_4$. Then P_3 is given by

- a) $P_3 = R_2^T R_1^T P_1 + R_2^T P_2 + R_2^T R_1^T P_4$
- b) $P_3 = -R_1^T R_2^T P_1 - R_2^T P_2 + R_1^T R_2^T P_4$
- c) $P_3 = -R_2^T R_1^T P_1 - R_2^T P_2 + R_2^T R_1^T P_4$
- d) $P_3 = R_1^T R_2^T P_1 - R_2^T P_2 + R_2^T R_1^T P_4$

Question-10

A rotation matrix is given by $R = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.25 & 0.433 & -0.866 \\ 0.43 & 0.75 & 0.5 \end{bmatrix}$. What is the corresponding unit quaternion in the form $q = \epsilon_1 i + \epsilon_2 j + \epsilon_3 k + \epsilon_4$ to 2 decimal places, where

$$\epsilon_4 = \frac{1}{2} \sqrt{1 + R_{11} + R_{22} + R_{33}}$$

$$\epsilon_1 = \frac{R_{32} - R_{23}}{4\epsilon_4}$$

$$\epsilon_2 = \frac{R_{13} - R_{31}}{4\epsilon_4}$$

$$\epsilon_3 = \frac{R_{21} - R_{12}}{4\epsilon_4}$$

, R_{ij} is the element in the rotation matrix at i -th row and j -th column.

- a) $q = -0.48i + 0.13j + 0.22k + 0.84$
- b) $q = 0.48i - 0.13j + 0.22k + 0.84$
- c) $q = -0.13i + 0.22j + 0.84k + 0.48$
- d) $q = 0.22i + 0.84j + 0.48k - 0.13$

Question-11

If the Jacobian matrix of a 3-link robot manipulator is $J = [J_1 \ J_2 \ J_3]$, where J_i , $i = 1, 2, 3$ are columns of the Jacobian matrix, what is **NOT a condition** for singularity? Explain your answer.

Choices:

- 1) J_1 is colinear with J_3
- 2) $J_2 = 1.5J_3$
- 3) $J_1^T J_2 = 0$
- 4) J_1 is parallel to J_2

Question-12

The homogeneous transformation matrix from the end point frame to the base frame of a 3-link manipulator is given by

$${}^0T_4 = \begin{bmatrix} \cos(\theta_1 + \theta_2 + \theta_3) & -\sin(\theta_1 + \theta_2 + \theta_3) & 0 & L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2 + \theta_3) & \cos(\theta_1 + \theta_2 + \theta_3) & 0 & L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (11)$$

The Jacobian matrix of the above manipulator is given by

a)

$$J = \begin{bmatrix} -L_2 \sin(\theta_1 + \theta_2) - L_1 \sin(\theta_1) & -L_2 \sin(\theta_1 + \theta_2) & 0 \\ L_2 \cos(\theta_1 + \theta_2) + L_1 \cos(\theta_1) & L_2 \cos(\theta_1 + \theta_2) & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (12)$$

b)

$$J = \begin{bmatrix} -L_2 \sin(\theta_1 + \theta_2) - L_1 \sin(\theta_1) & 0 & -L_2 \sin(\theta_1 + \theta_2) \\ L_2 \cos(\theta_1 + \theta_2) + L_1 \cos(\theta_1) & L_2 \cos(\theta_1 + \theta_2) & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (13)$$

c)

$$J = \begin{bmatrix} -L_2 \sin(\theta_1 + \theta_2) - L_1 \sin(\theta_1) & L_2 \cos(\theta_1 + \theta_2) + L_1 \cos(\theta_1) & 0 \\ -L_2 \sin(\theta_1 + \theta_2) & L_2 \cos(\theta_1 + \theta_2) & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (14)$$

Question-13

What would be the joint torques needed to exert a force vector $\mathbf{f} = [1 \ 1 \ 1]^T$ at the tip of the 3-link manipulator with the homogeneous transformation matrix:

$${}^0\mathbf{T}_4 = \begin{bmatrix} \cos(\theta_1 + \theta_2 + \theta_3) & -\sin(\theta_1 + \theta_2 + \theta_3) & 0 & L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2 + \theta_3) & \cos(\theta_1 + \theta_2 + \theta_3) & 0 & L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (15)$$

a)

$$\boldsymbol{\tau} = \begin{bmatrix} L_2 \sin(\theta_1 + \theta_2) + L_1 \sin(\theta_1) + L_2 \cos(\theta_1 + \theta_2) + L_1 \cos(\theta_1) \\ -L_2 \sin(\theta_1 + \theta_2) + L_2 \cos(\theta_1 + \theta_2) \\ 0 \end{bmatrix} \quad (16)$$

b)

$$\boldsymbol{\tau} = \begin{bmatrix} -L_2 \sin(\theta_1 + \theta_2) - L_1 \sin(\theta_1) + L_2 \cos(\theta_1 + \theta_2) + L_1 \cos(\theta_1) \\ -L_2 \sin(\theta_1 + \theta_2) + L_2 \cos(\theta_1 + \theta_2) \\ 0 \end{bmatrix} \quad (17)$$

c)

$$\boldsymbol{\tau} = \begin{bmatrix} -L_2 \sin(\theta_1 + \theta_2) - L_1 \sin(\theta_1) - L_2 \sin(\theta_1 + \theta_2) \\ L_2 \cos(\theta_1 + \theta_2) + L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) \\ 0 \end{bmatrix} \quad (18)$$

Question-14

What condition in joint angles in the 3-link manipulator with the homogeneous transformation matrix given by

$${}^0\mathbf{T}_4 = \begin{bmatrix} \cos(\theta_1 + \theta_2 + \theta_3) & -\sin(\theta_1 + \theta_2 + \theta_3) & 0 & L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2 + \theta_3) & \cos(\theta_1 + \theta_2 + \theta_3) & 0 & L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (19)$$

will result in a torque only around joint-1 to exert a force vector $\mathbf{f} = [1 \ 1 \ 1]^T$ at the tip?

a) $\theta_2 = 0$

b) $\theta_2 + \theta_3 = \pi/2$

c) $\theta_2 + \theta_3 = \pi/4$

d) $\theta_1 + \theta_2 = \pi/4$

Question-15

Which of the following vectors is in a homogeneous coordinate notation?

a)

$$\mathbf{v}_1 = [2 \ 4 \ 1 \ 0]^T \quad (20)$$

b)

$$\mathbf{v}_2 = \begin{bmatrix} 4 & 8 & 2 & 2 \end{bmatrix}^T \quad (21)$$

c)

$$\mathbf{v}_3 = \begin{bmatrix} 4i & 8j & 2k \end{bmatrix}^T \quad (22)$$

Question-16

- a) What is the purpose of the first rotation of a $Z - Y - X$ Euler rotation?
- b) Likewise, what is the purpose of the second rotation?

Question-17

How many sequences are there for Euler rotations? i.e. $Z - Y - X$, $X - Y - Z$, ...

- a) For moving frame rotations
- b) For fixed frame rotations